

Function fields on elliptic curves — Isogeny

Perplexity

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Introduction

The notion of a function field is fundamental in algebraic geometry and number theory. For elliptic curves, the function field encapsulates the algebraic functions that can be defined on the curve and serves as a foundational concept connecting geometry, algebra, and arithmetic.

1 Elliptic Curves

An *elliptic curve* E over a field k is commonly given by a Weierstrass equation of the form:

$$E : y^2 = x^3 + ax + b,$$

where $a, b \in k$ and the cubic polynomial on the right-hand side has distinct roots (non-zero discriminant) to ensure nonsingularity.

2 Definition of the Function Field of an Elliptic Curve

The *function field* of the elliptic curve E , denoted $k(E)$, consists of all rational functions defined on E . Formally, these are ratios of polynomials in the coordinates x and y that respect the curve equation. More precisely:

$$k(E) = \left\{ f(x, y) = \frac{P(x, y)}{Q(x, y)} \mid P, Q \in k[x, y], Q \neq 0, \text{ and } y^2 = x^3 + ax + b \right\},$$

where the relation $y^2 = x^3 + ax + b$ allows polynomial expressions to be simplified by replacing occurrences of y^2 .

This field $k(E)$ is an extension of the rational function field $k(x)$, obtained by adjoining y subject to the relation imposed by the curve:

$$k(E) \cong \frac{k(x)[y]}{(y^2 - x^3 - ax - b)}.$$

2.0.1 Note

- $k(x)[y]$ — The Polynomial Ring in y over the Field $k(x)$

- $k(x)$ is the field of rational functions in x over k . Its elements are fractions of polynomials in x :

$$\frac{p(x)}{q(x)}$$

$$p(x), q(x) \in k[x], q(x) \neq 0.$$

- $k(x)[y]$ is the ring of polynomials in the variable y but whose coefficients come from the field $k(x)$. So each element is:

$$\sum_{j=0}^n r_j(x) y^j$$

where each $r_j(x) \in k(x)$

- This is also a ring, but now the coefficients are rational functions in x , allowing division by polynomials in x .
- The ring $k(x)[y]$ is often used to describe algebraic relations involving y when x is treated as a parameter or already allowed to be inverted.

• $k[x, y]$ — The Polynomial Ring in Two Variables

- $k[x, y]$ denotes the ring of polynomials in two variables x and y with coefficients in a field k .
- Elements of $k[x, y]$ look like finite sums:

$$\sum_{i,j} a_{i,j} x^i y^j$$

where the coefficients $a_{i,j} \in k$, and $i, j \geq 0$ are non-negative integers.

- This is a ring, not a field: you can add and multiply polynomials, but division is not always possible within polynomials themselves unless dividing by a unit.
- It is used to represent algebraic equations, like the defining polynomial of an elliptic curve:

$$y^2 - (x^3 + ax + b) = 0,$$

which defines a subset (the curve) inside the affine plane over k .

- Ring/ Irreducible polynomial forms a field.

3 Algebraic Structure of Functions

Because of the relation $y^2 = x^3 + ax + b$, every element of the function field can be expressed as a linear combination of y with coefficients in $k(x)$:

$$f(x, y) = g(x) + h(x)y,$$

where $g(x), h(x) \in k(x)$ are rational functions in x .

Polynomials $P(x)$ and $Q(x)$

The numerator and denominator polynomials in these rational functions have standard polynomial form:

$$P(x) = \sum_{i=0}^n a_i x^i, \quad Q(x) = \sum_{j=0}^m b_j x^j,$$

where the coefficients a_i, b_j lie in the base field k , and exponents are non-negative integers. Powers of y greater than one are reduced via the curve equation to keep functions linear in y at most.

4 Operations in the Function Field

The set $k(E)$ forms a **field** under the operations of addition and multiplication of rational functions, defined analogously to fractions:

- **Addition:** For two functions

$$f = \frac{P}{Q}, \quad g = \frac{R}{S},$$

their sum is

$$f + g = \frac{P}{Q} + \frac{R}{S} = \frac{PS + RQ}{QS}.$$

- **Multiplication:** Their product is

$$f \cdot g = \frac{P}{Q} \cdot \frac{R}{S} = \frac{PR}{QS}.$$

In these operations, simplifications are performed by replacing y^2 with $x^3 + ax + b$, ensuring the functions remain consistent within the function field.

5 Intuition and Geometric Interpretation

Geometrically, the function field $k(E)$ represents all algebraic measurements and rational functions definable on the elliptic curve E . It captures the structure of the curve in algebraic terms and serves as the algebraic counterpart of the coordinate ring for affine varieties.

This function field plays an essential role in understanding morphisms between elliptic curves (such as isogenies), as these induce field homomorphisms between their respective function fields.

Summary

To summarize the interrelated concepts:

- An elliptic curve E is defined by a cubic equation relating x and y .
- The function field $k(E)$ consists of rational functions $f(x, y)$ that satisfy the curve equation.
- These functions can always be expressed as rational functions in x , linearly extended by y , with polynomial coefficients having non-negative integer exponents.

- Addition and multiplication in $k(E)$ behave like fraction operations, with reduction via the defining equation of E .
- The function field encodes the algebraic and geometric nature of the curve and is fundamental for studying mappings between elliptic curves.

6 Notes

1. can i say that $k(E) = \frac{k(x)[y]}{(y^2 - x^3 - ax - b)}$ forms a torus?

2.